

ILYA PAPOU

8 years in IT · 5+ in DevOps & SRE

Senior Platform & SRE Engineer with experience building and operating production Kubernetes platforms, automating software delivery, and leading DevOps teams. My background includes cloud and bare-metal infrastructure, security controls, observability, incident response, and disaster recovery. My current personal R&D focuses on agent infrastructure for software engineering, including MCP, A2A, and workflows with human review and approval.

PROFESSIONAL EXPERIENCE

TRAVEL TECH — SENIOR SRE & PLATFORM ENGINEER

Aviation & travel

Mar 2025 — Sep 2026 · Contract · Remote

- Built and operated a bare-metal Kubernetes platform using RKE2, with encrypted etcd, mTLS across services and data systems, and centralized certificate management through Vault.
- Standardized deployments with GitLab CI/CD, reusable Helm charts, and Argo CD. Integrated security checks into release workflows to block vulnerable builds and production deployments.
- Owned on-call incident response, investigated failures, and documented recovery procedures and preventive actions.
- Integrated kagent and Mattermost for AI-assisted engineering, managing agent definitions and platform configuration through IaC and GitOps.

INSURANCE BROKERAGE — HEAD OF DEVOPS

Finance & insurance

Jun 2023 — Mar 2025 · Full-time · Hybrid

- Built and led a DevOps team with four direct reports, coordinating an infrastructure group of ~10 people. Established onboarding, training, knowledge sharing, and regular technical reviews.
- Built cloud infrastructure from the ground up and provisioned ~8 Kubernetes clusters (20+ nodes total) with Terraform and GitOps. Improved availability through Multi-AZ and Multi-Region design, integrating cloud services with VMware infrastructure.
- Led service migration to the cloud with minimal downtime and controlled change management.
- Reduced infrastructure costs through automated shutdown of unused resources and usage governance.
- Built reusable Jenkins delivery pipelines for 30–40 microservices with Helm, security checks, tests, and approvals. Build and delivery took ~5 minutes for batches of more than six services.
- Introduced Vault, encrypted service communication, centralized monitoring, and tested disaster recovery procedures. Developed a Keycloak SSO proof of concept and coordinated requirements with security and development teams.

IT SERVICES COMPANY — SENIOR DEVOPS ENGINEER

IT services

Apr 2022 — Jun 2023 · Full-time · Remote

- Managed development and test infrastructure across approximately ten virtualization hosts, each running around 50 virtual machines.
- Deployed Kubernetes in an air-gapped Linux environment and automated offline releases.
- Maintained data, messaging, and delivery services, including PostgreSQL/Patroni, Cassandra, RabbitMQ, Airflow, TeamCity, and Nexus. Automated operations with Ansible, Python, and Bash.

FLANT & FREELANCE — SOFTWARE DEVELOPMENT & DEVOPS

Cross-industry IT consulting

Nov 2019 — Mar 2022 · Selected projects · Remote

- Delivered cloud infrastructure and web applications across freelance and Flant projects, including work with AWS services.
- Built React/Next.js and Node.js applications and deployed Java/Spring workloads to Kubernetes using Helm and automated delivery workflows.
- Supported delivery and operations with Prometheus, Grafana, ELK, and PostgreSQL/Patroni.

EARLIER EXPERIENCE

Information Security Intern · Telecommunications company · Feb 2018 — Nov 2019



CONTACT

Buenos Aires, Argentina

✉ ilya@papou.email

🌐 papou.work

📧 papou.work

✈ pilprod

📄 pilprod

🔑 pilprod

SPOKEN LANGUAGES

Russian: Native

English: B1

Spanish: A1

German: A1

CORE STRENGTHS

Platform ownership

Production reliability

Delivery automation

Team leadership

ENVIRONMENTS

Cloud & bare metal

Isolated, air-gapped

Product & consulting

Remote teams

DOMAIN EXPERIENCE

Finance & insurance

Aviation & travel

iGaming & anti-fraud



AGENT ORCHESTRATION INFRASTRUCTURE

Jun 2026 – Present · Personal R&D · PoCs

- Personal R&D platform for developers and AI coding agents working on software and infrastructure tasks. Designed to preserve context across agent handoffs, connect tools and long-running workflows, and keep changes under human review and approval.
- Deployed platform components on Google Cloud and built four reusable Helm workload profiles with configuration validation and automated tests.
 - Built a Go Agent Host for native and Docker execution, with adapters for official Codex and Claude clients. Implemented A2A streaming, session continuation and cancellation, with TLS and short-lived access for host connections.
 - Added declarative agent configuration, model selection and separate development and production environments. Prepared Kubernetes workload isolation and default-deny network policies.
 - Designed A2A task handoffs and long-running workflows with kagent and Temporal, including retries and Human-in-the-loop approval gates for code and infrastructure changes.
 - Implemented MCP tool allowlists and a Go policy API with strict request validation, deny-by-default decisions and audit records. Pinned runtime skills to immutable artifact digests.
 - Built and ran container-image and Helm-chart release workflows. Prepared Google Cloud scanning and approval-based promotion pipelines.
 - Tested release policies and validated Agent Host contracts with simulated Codex and Claude providers.

Code [Platform](#) [Agent runtime](#) [kagent fork](#) [kagent integration](#) [Mattermost fork](#) [Image build](#)

TECHNICAL SKILLS

Platform & delivery

Linux · Kubernetes · RKE2 · Docker
Terraform · HCP Terraform Stacks
Helm · Argo CD · Ansible
GitLab CI/CD · Jenkins
GitHub Actions · Nexus

Cloud infrastructure

AWS · GCP
S3 · Lambda · EKS · IAM
GKE · Cloud SQL · GCS
Artifact Registry · Cloud Build & Deploy
IAM · WIF · Secret Manager
Cloud KMS · Pub/Sub · Cloudflare

Software engineering

Go · Python · Bash · JavaScript
React · Next.js · Node.js · Express
Java · Spring · C++
OpenAPI · gRPC
Schema validation · contract tests

Security & reliability

Vault · ESO · Keycloak · mTLS · Kyverno
GitGuardian · Trivy
Prometheus · VictoriaMetrics
Grafana · ELK · OpenTelemetry
Tailscale · OPNsense

ML & data infrastructure

Airflow · Slurm · JupyterHub · MLflow
PostgreSQL · Patroni · Redis
Kafka · RabbitMQ · Cassandra
MinIO · VectorDB · BigQuery

Agent infrastructure & R&D

kagent · agentgateway
MCP · A2A · RAG
Temporal · Human-in-the-loop
LangChain · ADK
Codex · Claude Code · Gemini
vLLM · Langfuse

HOME AEROPONICS & IOT AUTOMATION

Aug 2024 – Jan 2025 · Personal R&D

- Designed and built an integrated aeroponic system: drew wiring schematics, soldered electronics, and connected ESP32, Arduino, Raspberry Pi, sensors and actuators.
- Developed C++ sensor firmware and Python automation for climate control, mist generation, water supply and circulation, tank-to-tank transfers and leak detection.
- Automated lighting and collected light-quality data to guide adjustments. Used computer vision to assess experimental results through changes in leaves.
- Connected ESP and Zigbee devices through Mosquitto and Zigbee2MQTT using JSON over MQTT, with Home Assistant as the control, monitoring and alerting interface.
- Automated Raspberry Pi setup with Ansible, prototyped pH, TDS and EC monitoring, and began preparing Prometheus metrics collection.

Code [Controllers](#) [Sensor firmware](#)

ZERO-TRUST MESH & OPEN-SOURCE NGFW

Jan 2025 – Mar 2025 · Personal R&D

- Built an encrypted, multi-region Tailscale lab mesh linking devices and private networks across cloud, bare metal and homelabs.
- Automated network configuration and enabled one-click onboarding of new nodes.
- Managed all mesh ACLs through GitOps, with distinct user and service-account permissions for DNS discovery and service access.
- Tested OPNsense as an NGFW for subnet routing, network segmentation and traffic filtering.

Code [Ansible](#) [Access policy](#) [GCP network lab](#)